GEOG 2475: Geographic Information Systems I

Lecture 02 Datum and Projection

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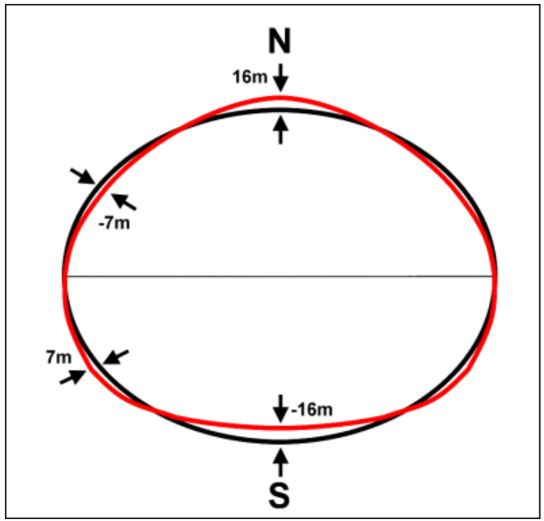
Overview

- Datum
  - ✓ Shape of Earth
  - ✓ Geoid
- Coordinate system
  - ✓ Geographic coordinate systems
  - ✓ Projected coordinate systems
- Projections
  - ✓ Plane projection
  - ✓ Conic projection
  - ✓ Cylindrical projection
  - ✓ Examples: Global, continental, and state scales

Shape of the Earth

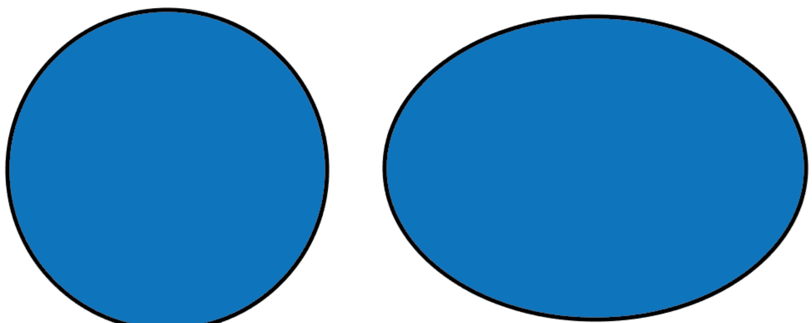
- Geodesy
  - ✓ Science of measuring Earth's shape



https://commons.wikimedia.org/wiki/File:Geoundnsrp.png

Shape of the Earth

- Oblate spheroid
  - ✓ Equator has larger circumference than meridians
  - ✓ Mass not evenly distributed



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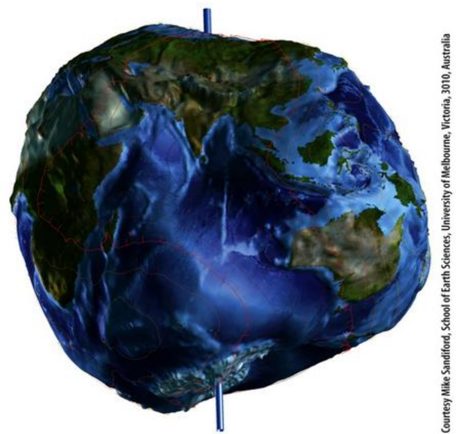


Figure 1.2

Modeling the Earth

True Earth surface

↓

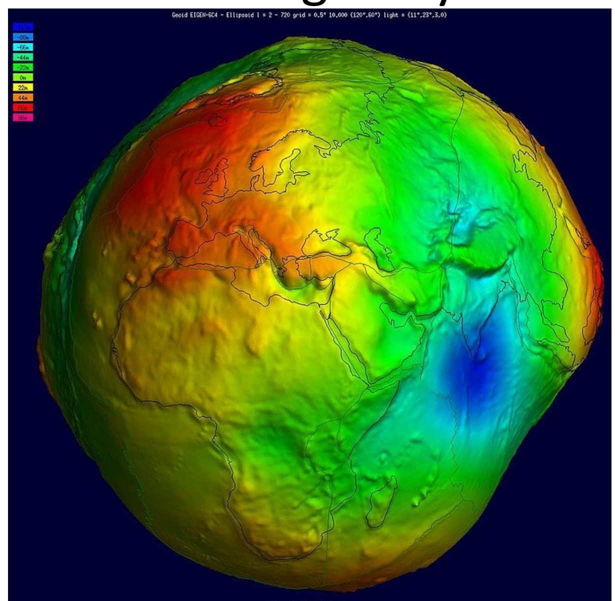
Geoid (Includes ocean)

↓

Ellipsoid

Geoid

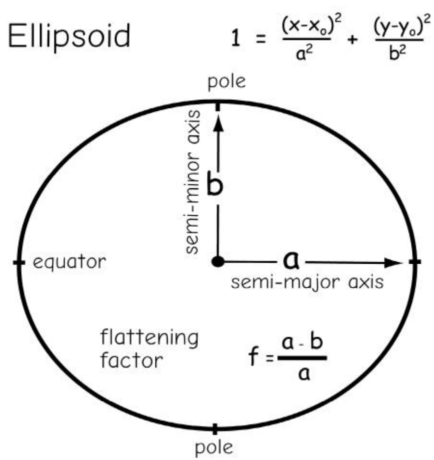
The geoidal surface can be thought of as an imaginary sea that covers the entire Earth and is not affected by wind, waves, the Moon, or forces other than Earth’s gravity.

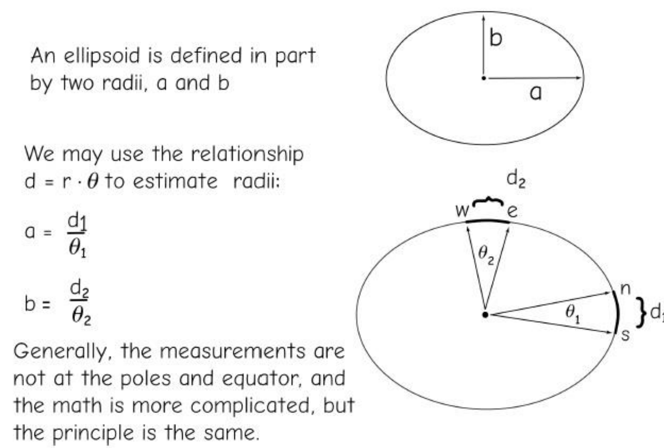


https://en.wikipedia.org/wiki/Geoid

Datum

- We need a mathematically simple model of the Earth's Surface
- We use an ellipsoid/ spheroid
- There are different ellipsoids/ spheroids used
- Combined with ground survey data, we call these datums





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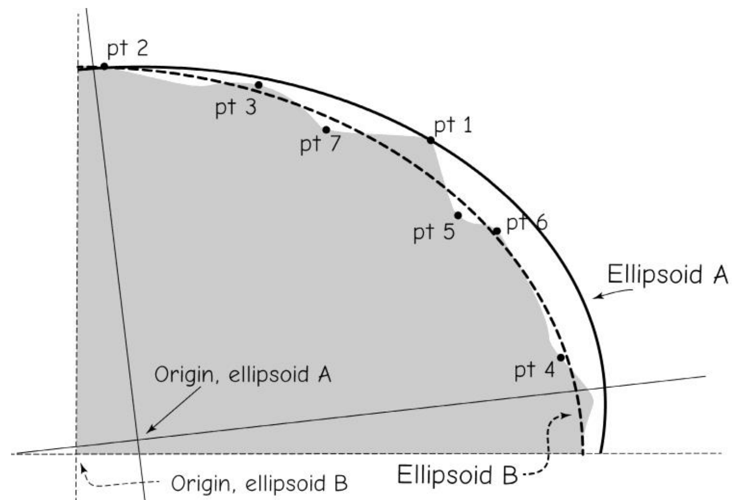
Datum

- Datum changes through time due to more survey measurements available or improvement on measurement methods
  - ✓ Before satellite era and after satellite era
  - ✓ Datum adjustment



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Datum

- **NAD27:** North American Datum of 1927
- **NAD83:** North American Datum of 1983
  - ✓ NAD83(xx), xx represents year, version of datum adjustment
  - ✓ NAD83(HARN/HPGN): High Accuracy Reference Networks or High Precision Geodetic Networks
- **WGS84:** World Geodetic System 0f 1984
  - ✓ WGS84(xx)
- **ITRF:** International Terrestrial Reference Frames
  - ✓ ITRF89, ITRF90, ITRF91

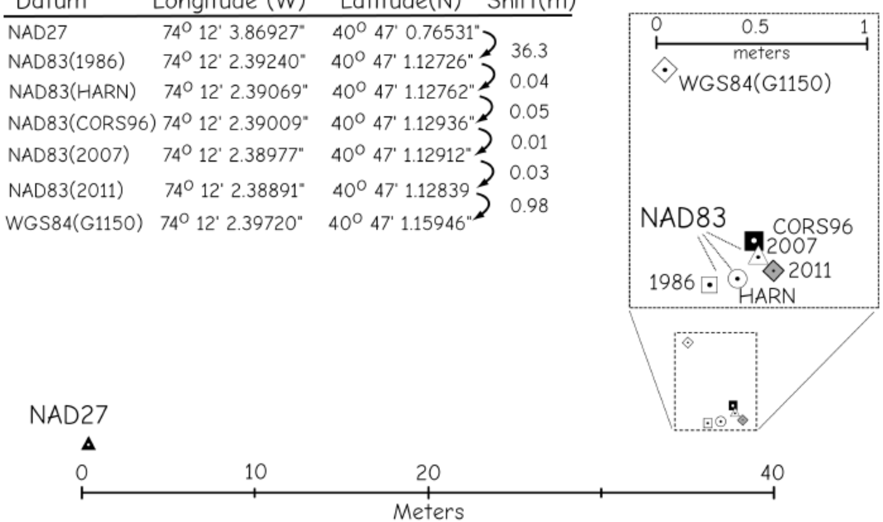
Datum

- WGS84 and ITRF can be considered equivalent for most purpose, as difference between them since 1995 are generally only a few centimetres

Examples of Datum Shifts

Successive datum transformations for New Jersey control point, Bloom 1

Datum	Longitude (W)	Latitude(N)	Shift(m)
NAD27	74° 12' 3.86927"	40° 47' 0.76531"	
NAD83(1986)	74° 12' 2.39240"	40° 47' 1.12726"	36.3
NAD83(HARN)	74° 12' 2.39069"	40° 47' 1.12762"	0.04
NAD83(CORS96)	74° 12' 2.39009"	40° 47' 1.12936"	0.05
NAD83(2007)	74° 12' 2.38977"	40° 47' 1.12912"	0.01
NAD83(2011)	74° 12' 2.38891"	40° 47' 1.12839	0.03
WGS84(G1150)	74° 12' 2.39720"	40° 47' 1.15946"	0.98



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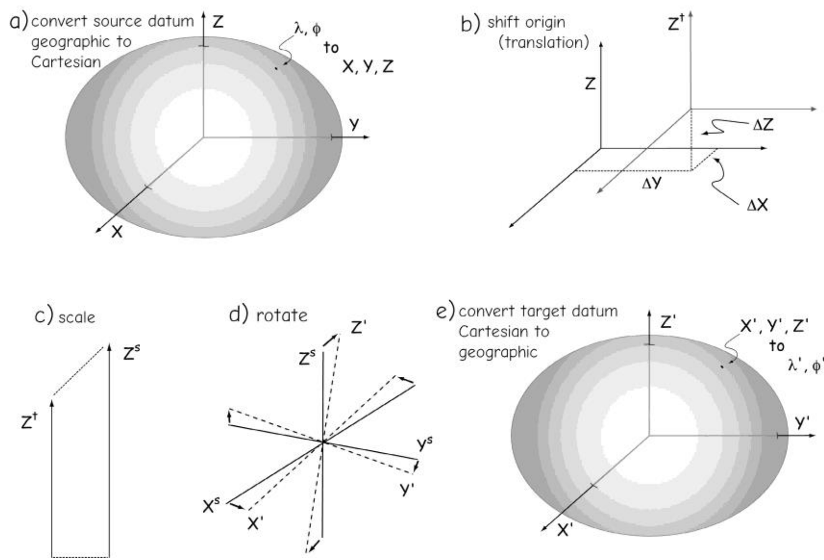
Datum

- Transformation
  - ✓ Converting coordinates from one datum to another
    - Shift origin
    - Scale
    - rotation



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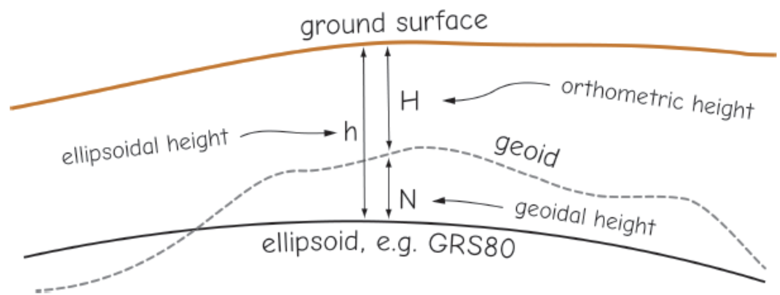
Datum

- Orthometric Height (H)
  - ✓ Height above geoid
- Ellipsoidal Height (h)
  - ✓ height above ellipsoid
- Geoid Height (N)
  - ✓ Geoid above ellipsoid

$$h = H + N$$

ellipsoidal height = orthometric height + geoidal height

h = H + N



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Datum

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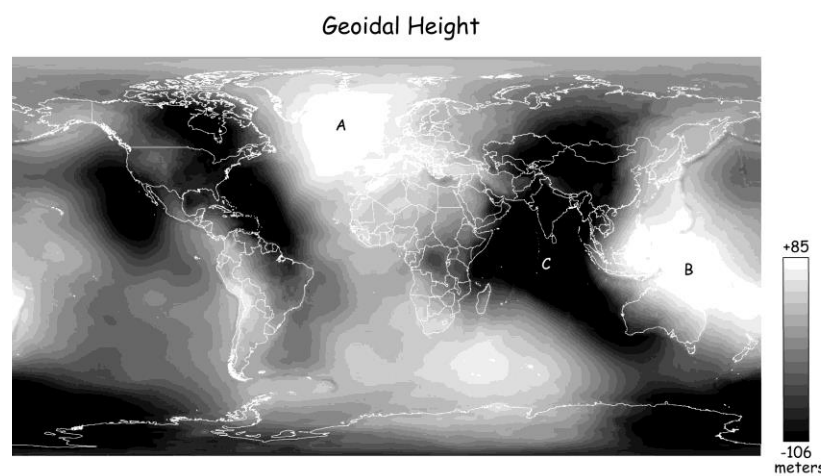


Common vertical datum

- NGVD29: National Geodetic Vertical Datum of 1929
  - ✓ No geoid, mean sea level
- NAVD88: North American Vertical Datum of 1988 (NAVD88)
  - ✓ GEOID03, GEOID09, GEOID12B

Common vertical datum

- Pairing for horizontal and vertical datums for North America

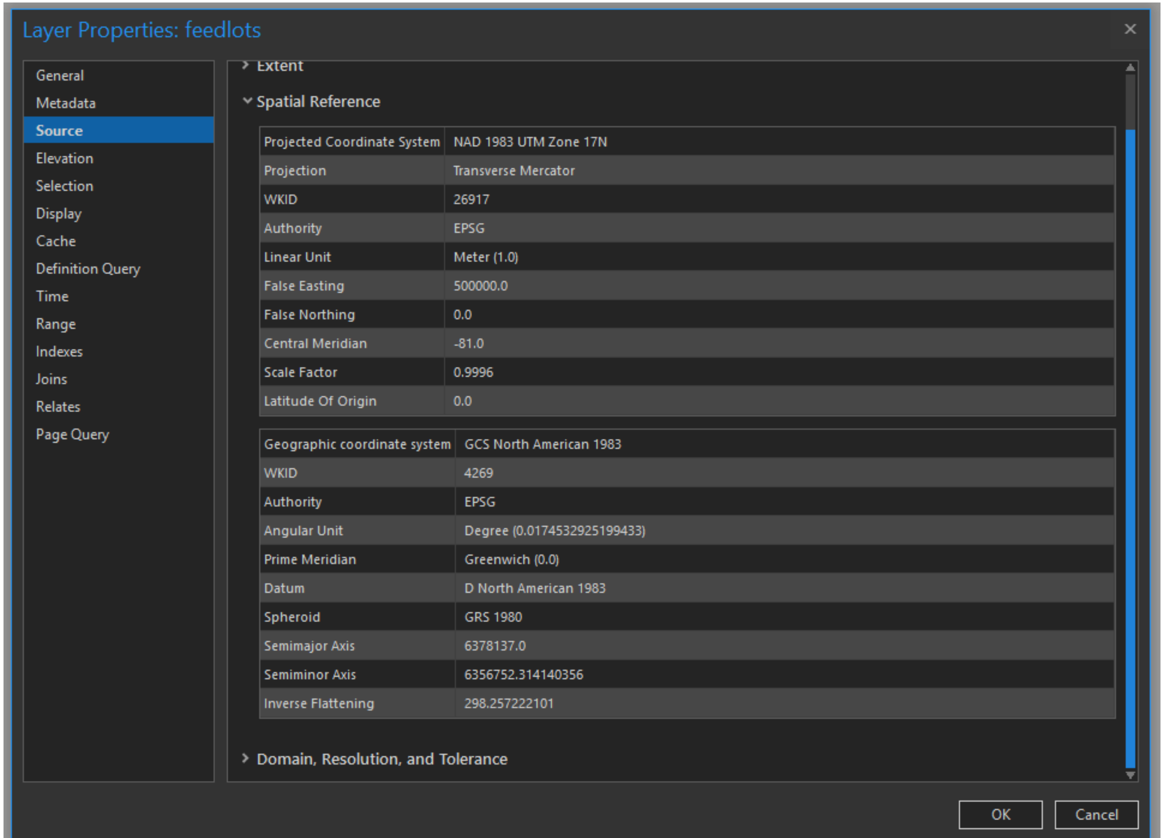
NGVD29, no geoid
with
NAD27, NAD83(1986)

NAVD88, Geoid 03
with
NAD83(1996)

NAVD88, Geoid 09
with
NAD83(NSRS2007)

NAVD88, Geoid12B
with
NAD83(2011)

Datum/Projection Info in ArcGIS



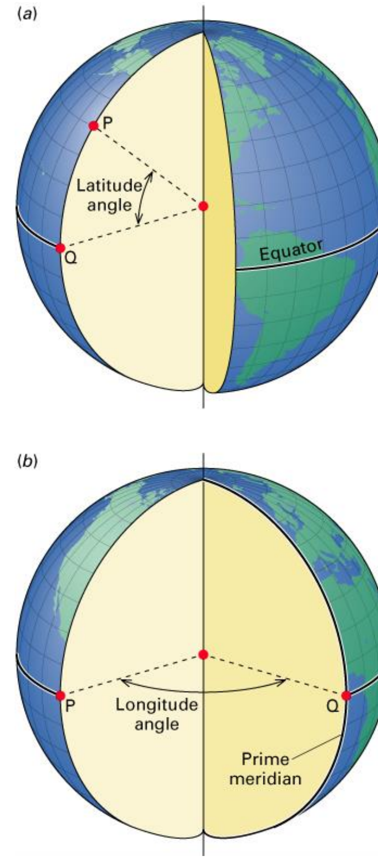
Projected Coordinate System	NAD 1983 UTM Zone 17N
Projection	Transverse Mercator
WKID	26917
Authority	EPSG
Linear Unit	Meter (1.0)
False Easting	500000.0
False Northing	0.0
Central Meridian	-81.0
Scale Factor	0.9996
Latitude Of Origin	0.0

Geographic coordinate system	GCS North American 1983
WKID	4269
Authority	EPSG
Angular Unit	Degree (0.0174532925199433)
Prime Meridian	Greenwich (0.0)
Datum	D North American 1983
Spheroid	GRS 1980
Semimajor Axis	6378137.0
Semiminor Axis	6356752.314140356
Inverse Flattening	298.257222101

ArcPro

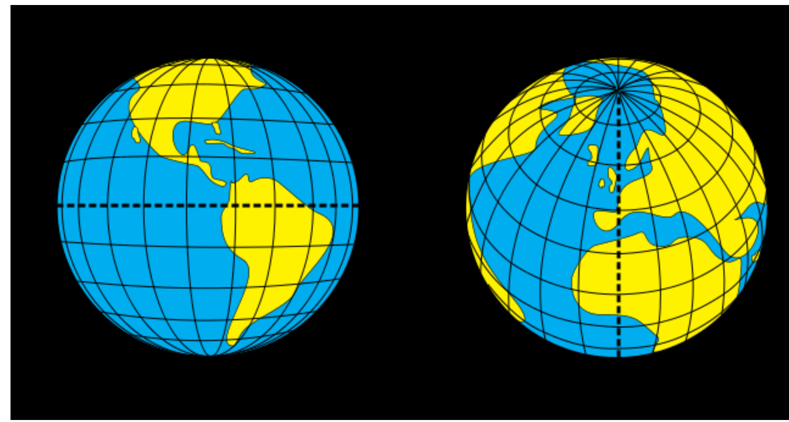
Geographic Coordinate Systems

- Coordinates on a Sphere
  - ✓ Latitude and Longitude
  - ✓ Specified as angles or degrees
- Latitude: Angles from equator
  - ✓ Range 0-90 N/S
- Longitude: Angles from prime meridian
  - ✓ Range 0-180 E/W



Geographic Coordinate Systems

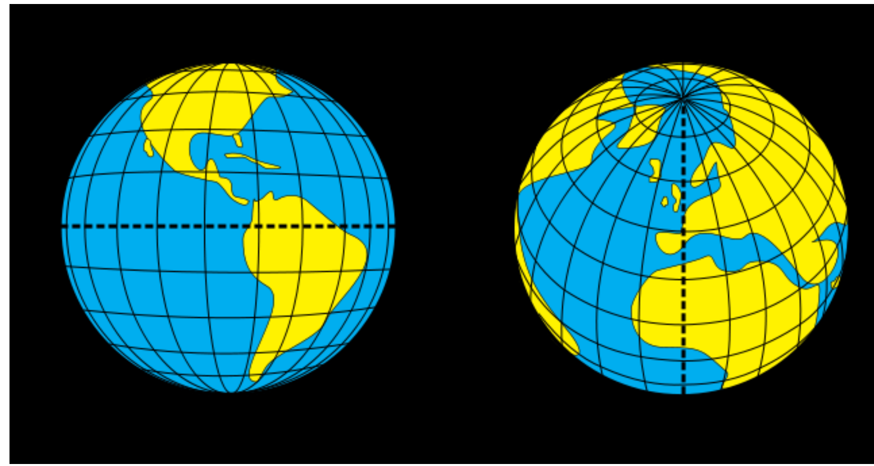
- Latitude (Parallels):
  - ✓ Measures north to south
  - ✓ 90° in each hemisphere
  - ✓ Increases as you go north in the northern hemisphere
  - ✓ Increases as you go south in the southern hemisphere
  - ✓ Zero in Equator



https://commons.wikimedia.org/wiki/File:Latitude_and_Longitude_of_the_Earth.svg

Geographic Coordinate Systems

- Longitude (Meridians)
  - ✓ Measures east to west
  - ✓ 180° in each hemisphere
  - ✓ Increases as you go east in the eastern hemisphere
  - ✓ Increases as you go west in the western hemisphere
  - ✓ Zero is Prime Meridian, going through Greenwich, England



https://commons.wikimedia.org/wiki/File:Latitude_and_Longitude_of_the_Earth.svg

Geographic Coordinate Systems

- 1 Degree = 60 Minutes = 3600 Seconds
  - ✓ ° = Degrees
  - ✓ ‘ = Minutes
  - ✓ “ = Seconds

  1.5 ° =1 ° 0.5X60’= 1 ° 30’

  1 ° 30’ = 1° + (30’/60’) = 1.5°

Geographic Coordinate Systems

- Advantages:
  - ✓ Accurate representation
  - ✓ Latitude and longitude lines
  - ✓ Without distortion
- Disadvantages
  - ✓ Expensive to make
  - ✓ Cumbersome to handle and store
  - ✓ Difficult to measure distance
  - ✓ Not fully visible at one view

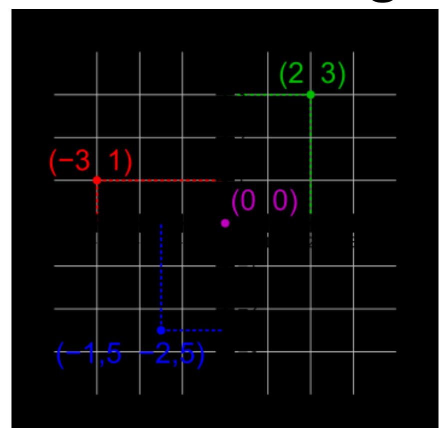
Image from ArcGIS API for JavaScript

Class Activity 2-1

- Explore the NGS Coordinate Conversion and Transformation Tool (NCAT) available from the U.S. NOAA/NGS website (https://www.ngs.noaa.gov/NCAT/index.xhtml;jsessionid=9190FFFE418DF889CAB0942A91BA7B9F) to convert the locations between different horizontal datums. And discuss:
  - ✓ What are your converted lats/lons?
  - ✓ Are the locations different with different datums?
  - ✓ Convert locations from NAD27 to NAD83(86)
    - Wisconsin: lat: 43°07’59”N and lon: 89°20’11’’W
    - Colorado: lat 40°00’00’’N and lon: 105°16’01’’ W
  - ✓ Convert locations from NAD83(86) to NAD27
    - Wash. D.C.: lat: 38°51’10.4052’N and lon: 77°02’19.9165’’ W

Projected Coordinate Systems

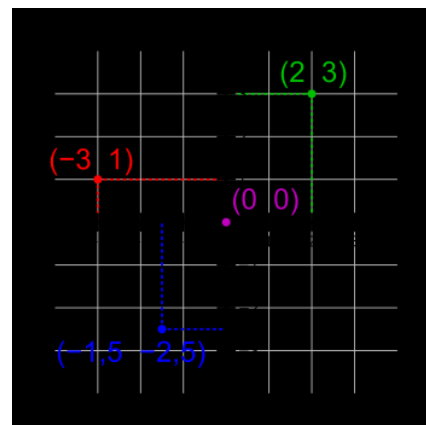
- Map projections allow us to view the 3D Earth on a 2D surface
- Earth is mathematically projected to a surface that can be made flat
- We now make measurements in distance or length units as opposed to degrees



https://commons.wikimedia.org/wiki/File:Cartesian_coordinate_system_(comma).svg

Projected Coordinate Systems

- They allow us to make measurements in a Cartesian coordinate system on a plane
- These 2D surface are known as map projections.



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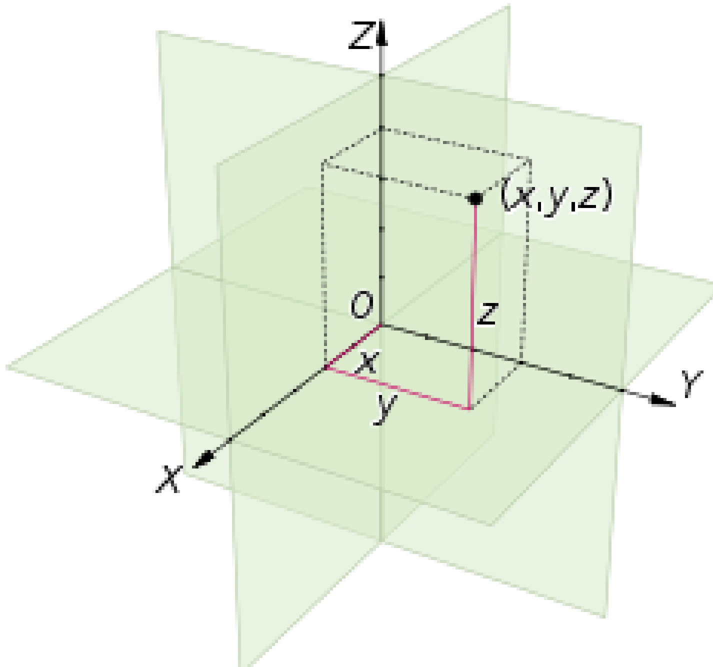
Projected Coordinate Systems

- X, Y, Z



https://en.wikipedia.org/wiki/Cartesian_coordinate_system

Geographic vs Projected Coordinate Systems

- Conversion from geographic to projected coordinates

Conversion from geographic (lon, lat) to projected coordinates

Given longitude = λ, latitude = ϕ (all angles in radians)

Mercator projection coordinates are:

$$x = R \cdot (\lambda - \lambda_o)$$
$$y = R \cdot \ln(\tan(\pi/4 + \phi/2))$$

where R is the radius of the sphere at map scale (e.g., Earth's radius), ln is the natural log function, and λ_o is the longitudinal origin (Greenwich meridian)

Inverse equation, from x, y to λ, ϕ:

$$\lambda = x/R + \lambda_o$$
$$\phi = (\pi/2) - 2 \cdot \tan^{-1}[e^{-y/R}]$$

Geographic vs Projected Coordinate Systems

Image from ArcGIS API for JavaScript

Image from ArcGIS API for JavaScript

Map projection

How?

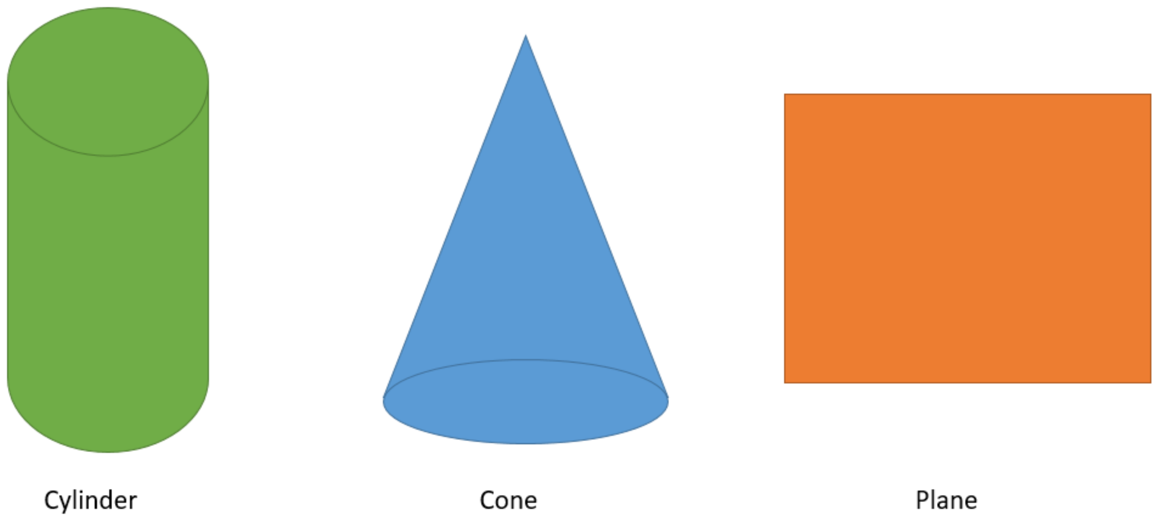
Shine a light from center of earth

Onto a "physical surface:"

1. Plane
2. Cone
3. Cylinder

Map projection

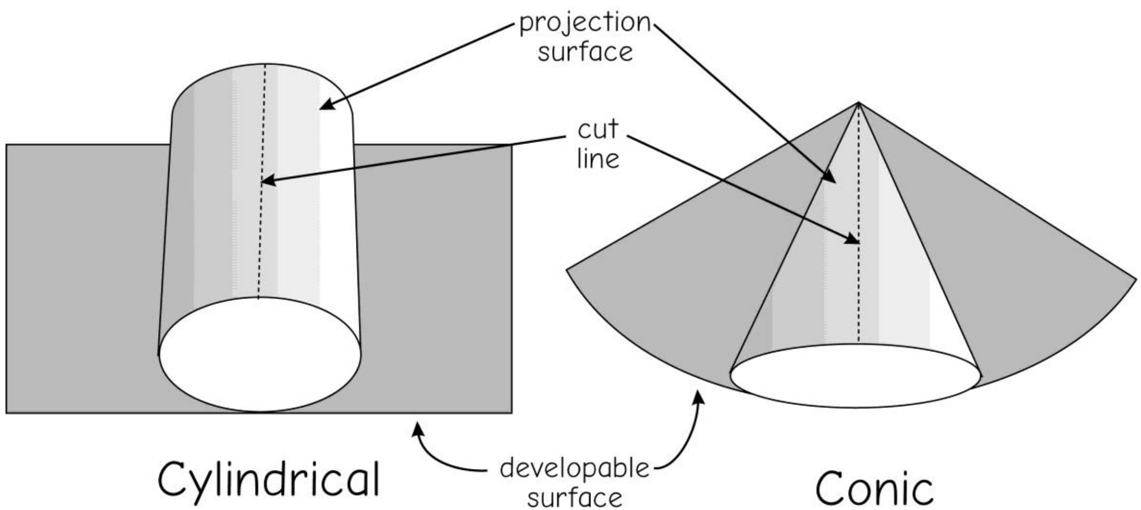
- Developable Surfaces
  - ✓ Geometric surface that can be flattened



Cylindrical Projections = Cylinder　Conic Projections = Cone　Azimuthal Projections = Plane

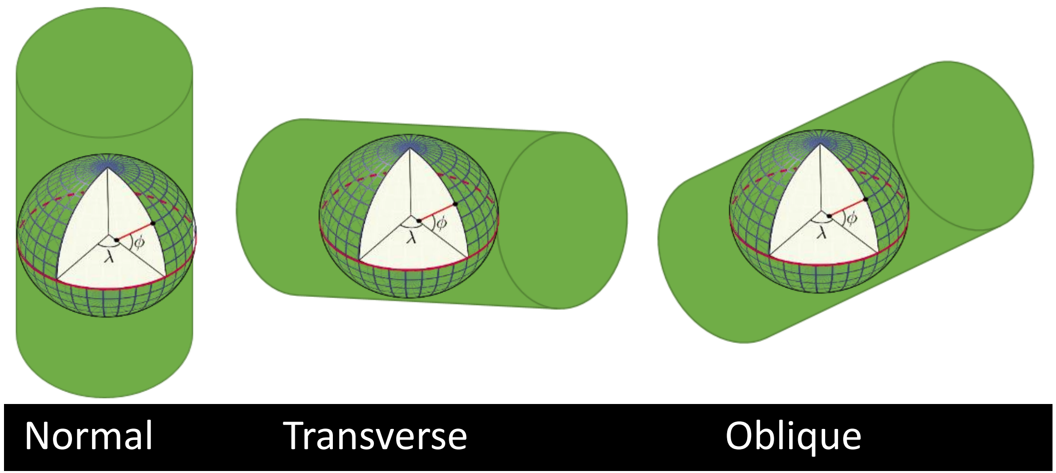
Map projection

- Developable Surfaces
  - ✓ Geometric surface that can be flattened



Map projection

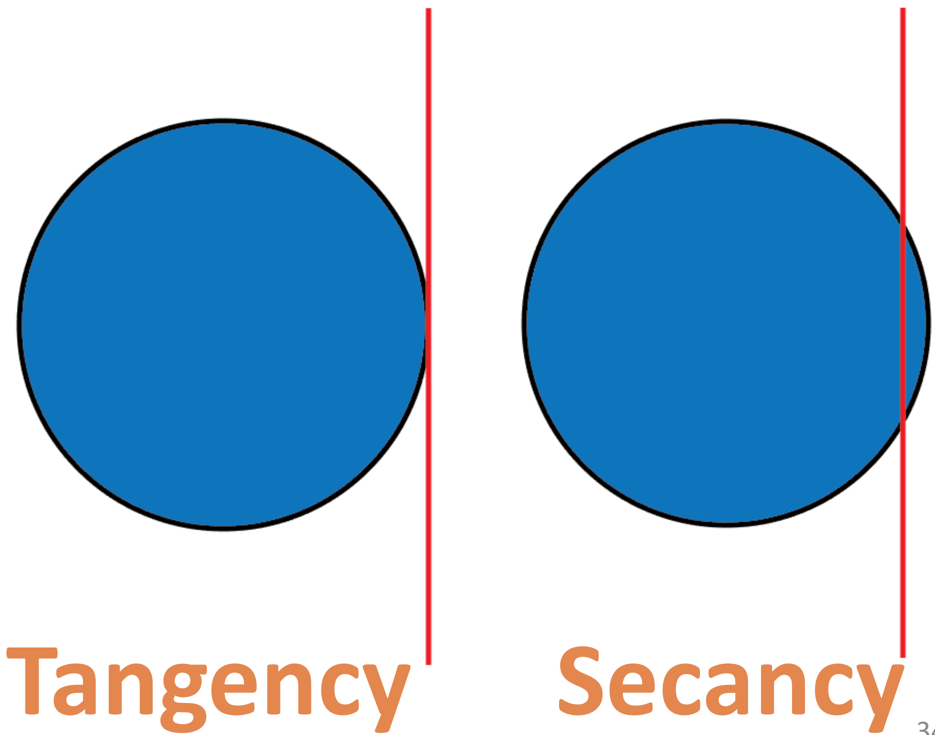
- Developable Surfaces
  - ✓ Different orientation of developable surface with Earth



https://commons.wikimedia.org/wiki/File:Latitude_and_longitude_graticule_on_an_ellipsoid.svg

Map projection

- Developable Surfaces
  - ✓ **Standard Line:** where projection touches the globe
    - **Standard Parallel:** standard line on a parallel
    - **Standard Meridian:** standard line on a meridian



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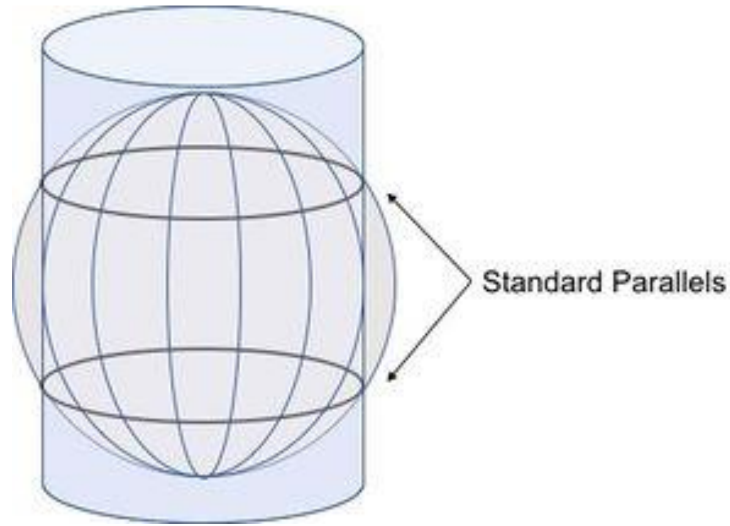
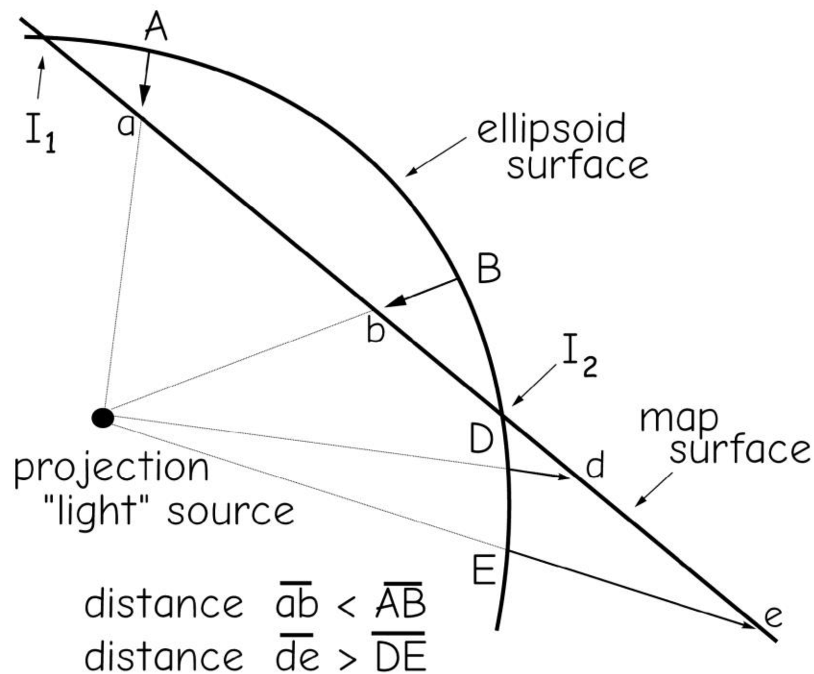


http://wiki.gis.com/wiki/index.php/Standard_parallel#:~:text=Standard%20parallels%20are%20parallel%20lines,is%20only%20one%20standard%20parallel.

Map projection

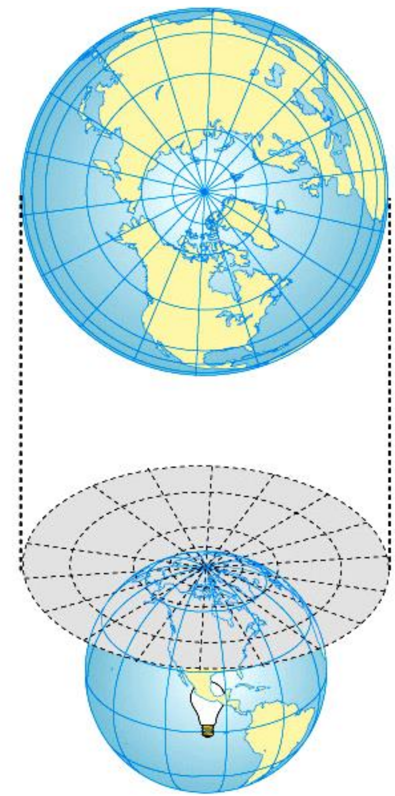
- Developable Surfaces
  - ✓ **Standard Line:** where projection touches the globe
    - No distortions at these locations! Distortions tend to increase with distance from the standards.



Map projection

○ Plane projection

- ✓ Plane positioned at pole
- ✓ **Grid:**
  - **Longitude:** straight, radiating
  - **Latitude:** circles
- ✓ Distortion increases away from center ("principal point")
- ✓ Good for polar regions
- ✓ Can't show more than half the world.



Map projection

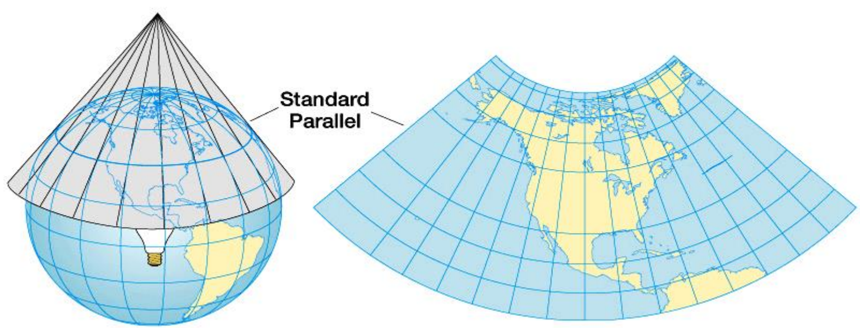
- Conic projection

✓ Hold cone over pole

✓ **Grid:**

  - **Longitude:** straight, converge
  - **Latitude:** curve
  - Distortion increases away from standard parallel

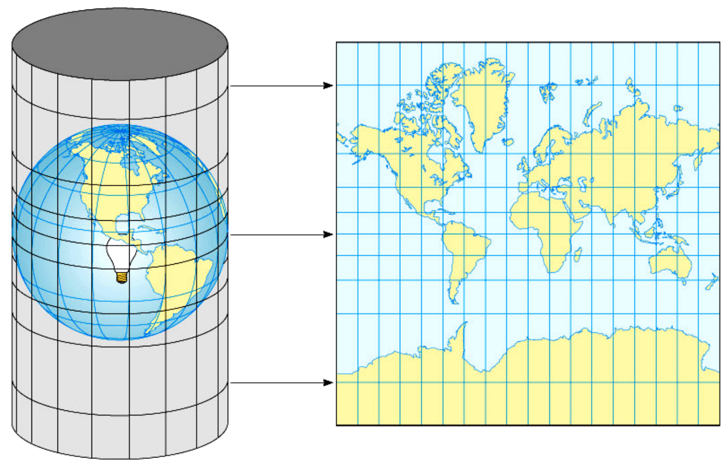
✓ Good for Mid-latitudes



Map projection

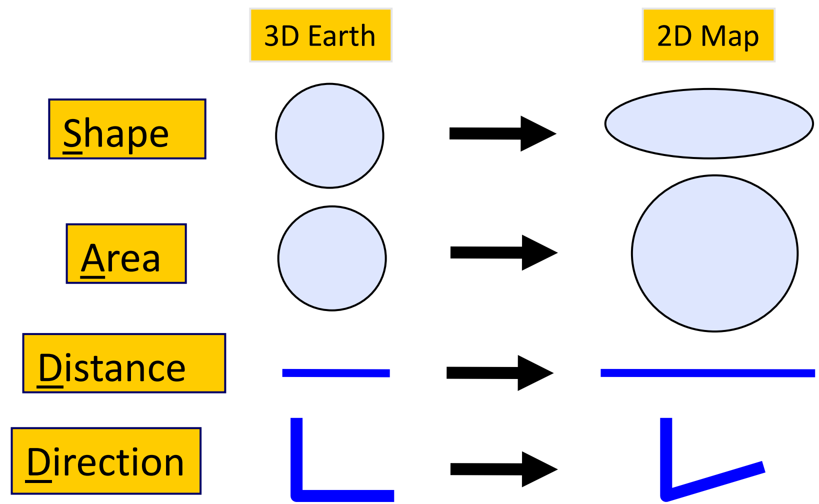
○ Cylindrical projection

- ✓ Cylinder around the equator
- ✓ Latitude & longitude: straight, intersect at 90°
- ✓ Distortion greatest at poles
- ✓ Good for low-latitude areas
- ✓ Good for navigation
- ✓ Mercator’s Projection



Map projection

- All map projections produce distortions of at least one of the following:
  - ✓ Shape
  - ✓ Area
  - ✓ Distance
  - ✓ Direction



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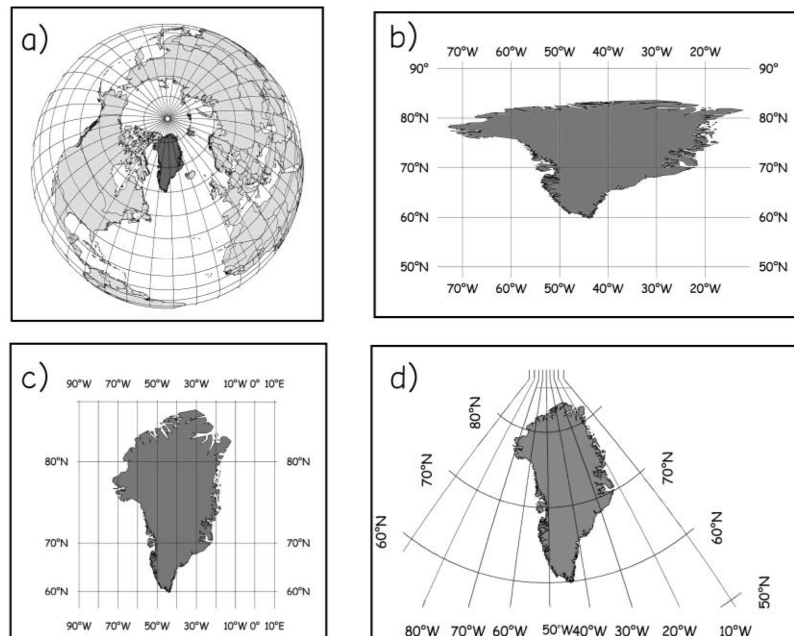


Mercator

Transverse Mercator

Bolstad, 2019

Class Activity 2-2

- Explore the true size tool https://thetruesize.com, discuss with your group members:
  - ✓ Do you know there are some distortions in the map before the class?
  - ✓ What did surprise you?

Map projection

Why all world maps are wrong?

Map projection

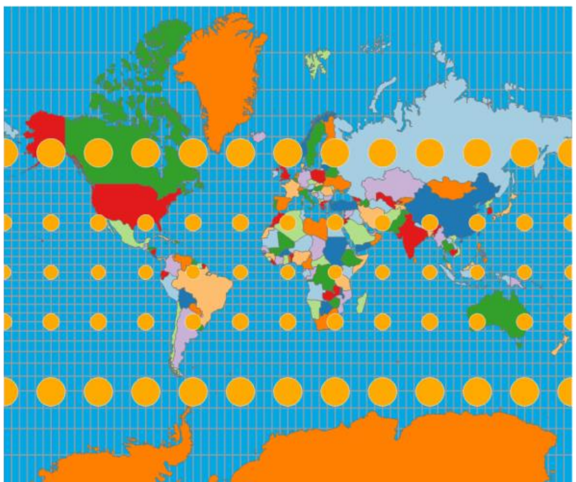
- Type
  - ✓ **Conformal Projections:** preserves local angles and shape
  - ✓ **Equivalent Projections:** represents areas in correct relative size
  - ✓ **Equidistant Projections:** maintains consistent scale or distance along certain lines
  - ✓ **Azimuthal Projections:** maintains certain accurate directions relative to certain vantage point

- Conformal and equivalent projections' properties are mutually exclusive
- Conformal and equivalent projections' properties are global properties

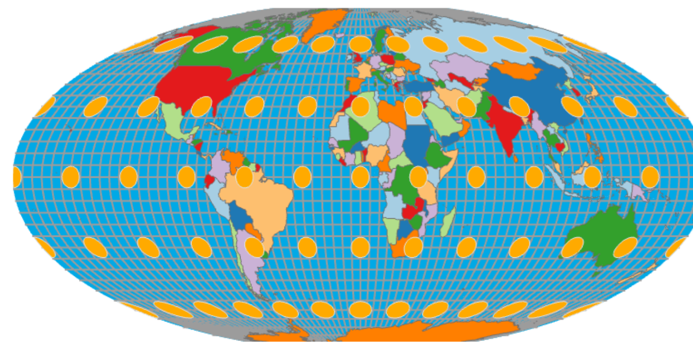
- Equidistant and azimuthal projections' properties are local properties and may be true only from or to the center of the map projection

Map projection

- Tissot Indicatrix
  - ✓ Without any distortions, the circles will all be the same shape, perfect circles, and size



Mercator projection
Distort size but not shape



Mollweide projection
Distort shape but not size

Map projection

- Examples

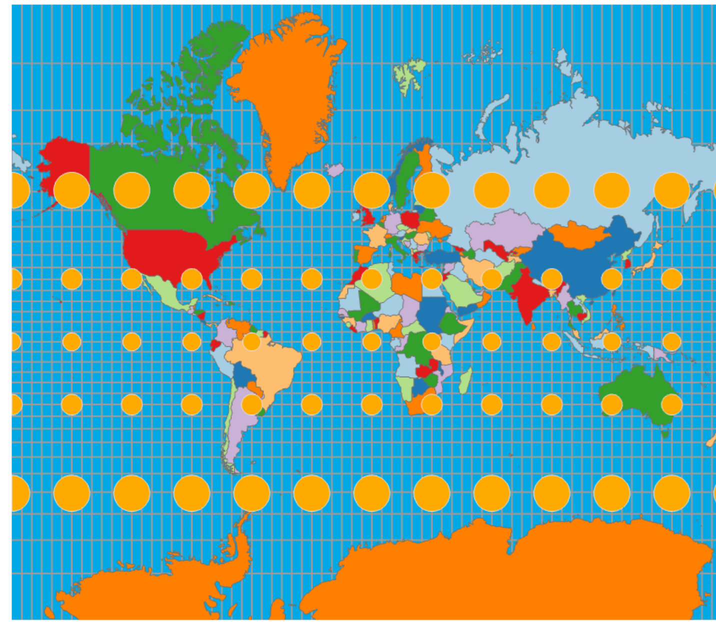
  ✓ Projected coordinate systems do not use longitude and latitude to measure locations

  ✓ Instead, they use map units of the projection itself:

    Meters (UTM, Albers, most others)
    Feet (State Plane, few others)

Map projection

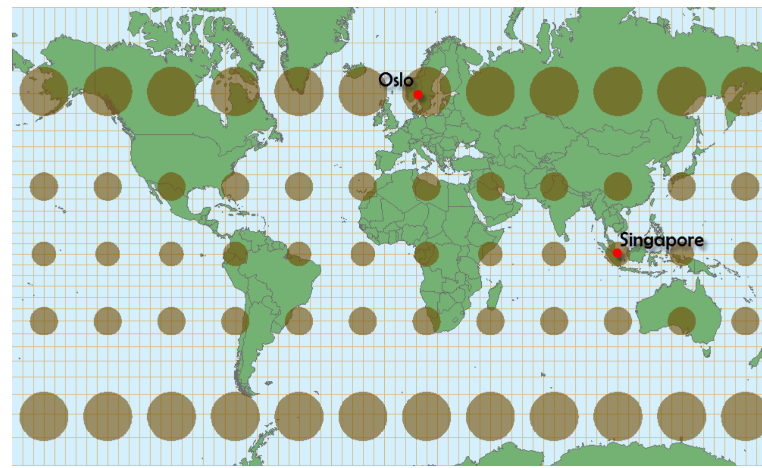
- World Projections: Mercator
  - ✓Cylindrical conformal projection
  - ✓Maintains shape
  - ✓Distorts size
  - ✓Parallels and meridians cross at right-angles



Tissot Indicatrix

Map projection

- World Projections: Web Mercator
  - ✓ Modification of Mercator projection that is commonly used for web maps
  - ✓ Bing Maps, OpenStreetMap, Google Maps, MapQuest, ArcGIS Online
  - ✓ Uses a sphere as opposed to a spheroid

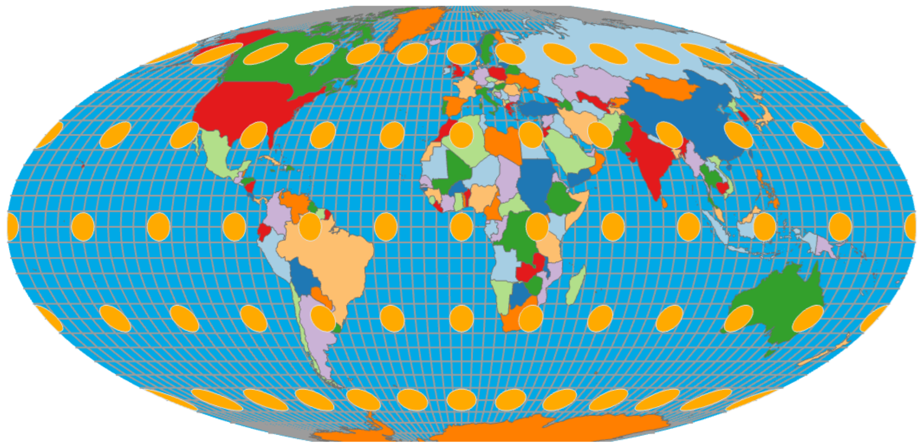


http://cartonerd.blogspot.com/2014/08/web-mercator-and-comparisons.html

Map projection

○ World Projections: Mollweide

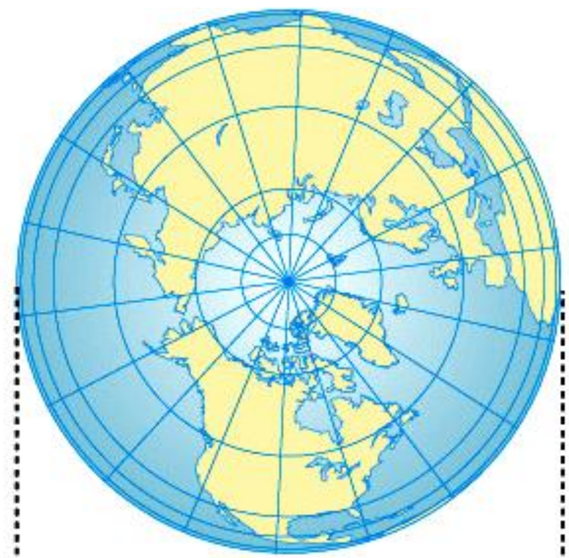
- ✓ Equivalent projection
- ✓ Maintains area
- ✓ Distorts shape and angle
- ✓ Pseudocylindrical
- ✓ Not based on developable surface



Map projection

○ World Projections: Azimuthal Equidistant

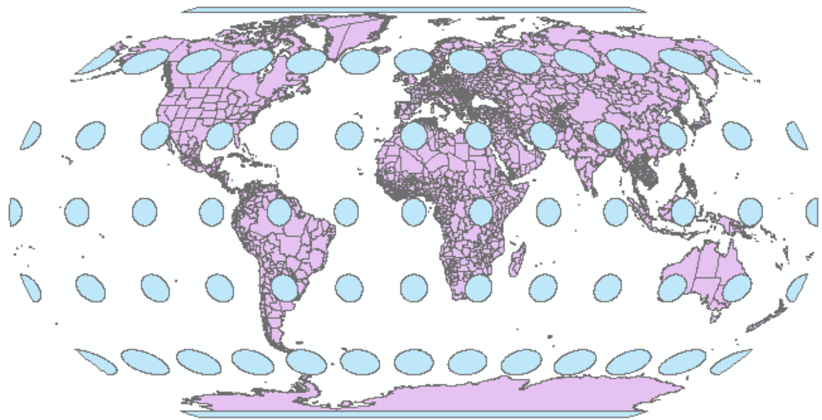
- ✓ Azimuthal and Equidistant
- ✓ Correct distance from center of map
- ✓ Correct angle from center of map
- ✓ Distorts shape and relative size



Map projection

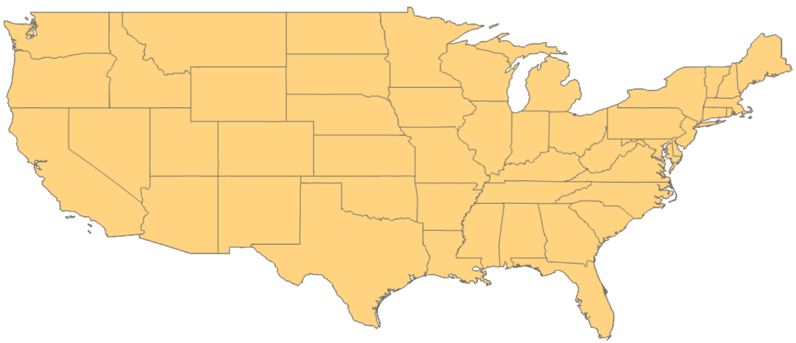
World Projections: Robinson

- ✓ Neither equivalent or conformal
- ✓ Strikes a balance between shape and size
- ✓ Meridians curve gently

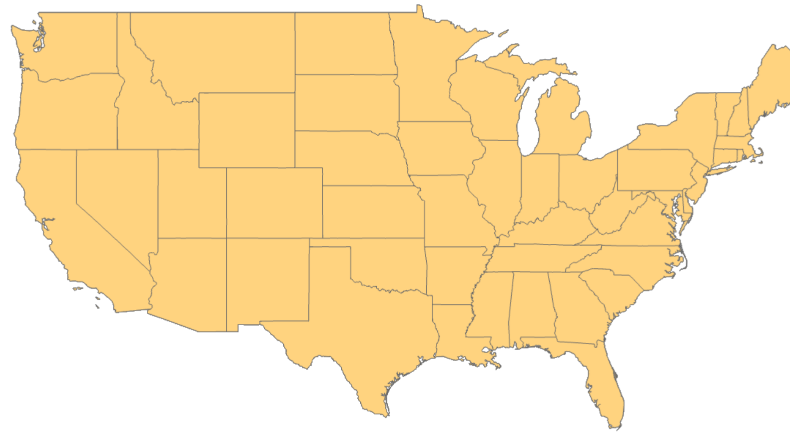


Map projection

- US projections



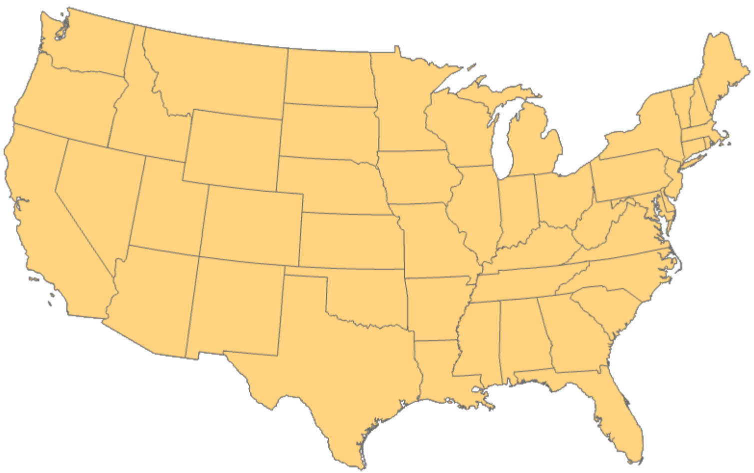
Not Projected



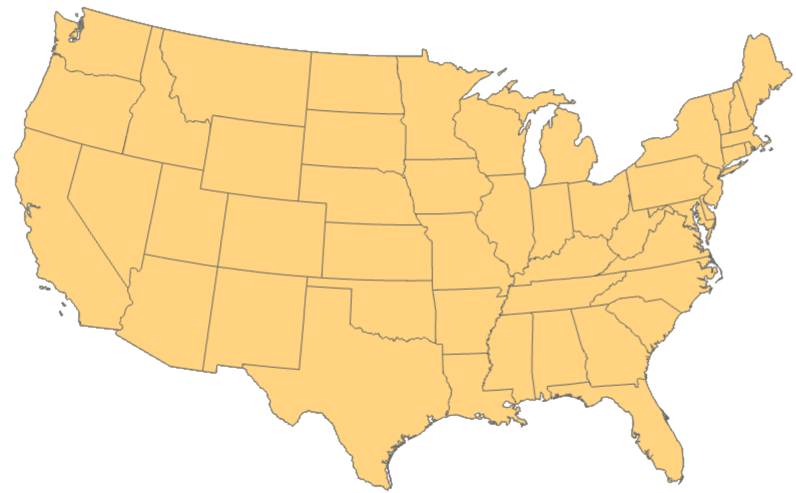
Mercator

Map projection

- US projections



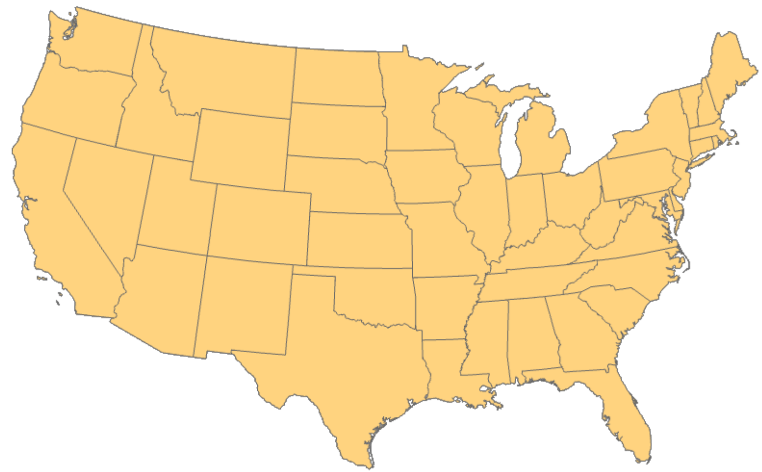
Albers Equal Area



Lambert Conformal Conic

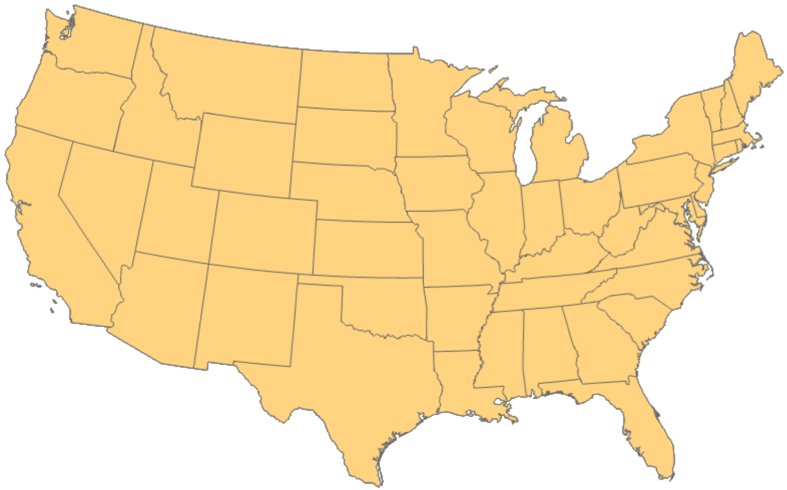
Map projection

- US projections: Albers Equal Area
  - ✓Conic
  - ✓Equivalent
  - ✓Maintains area
  - ✓Distort shape and local angles
  - ✓Two standard parallels



Map projection

- US projections: Lambert Conformal Conic
  - ✓ Conic
  - ✓ Conformal
  - ✓ Maintains shape and local angles
  - ✓ Distorts local size
  - ✓ Two standard parallels



Map projection

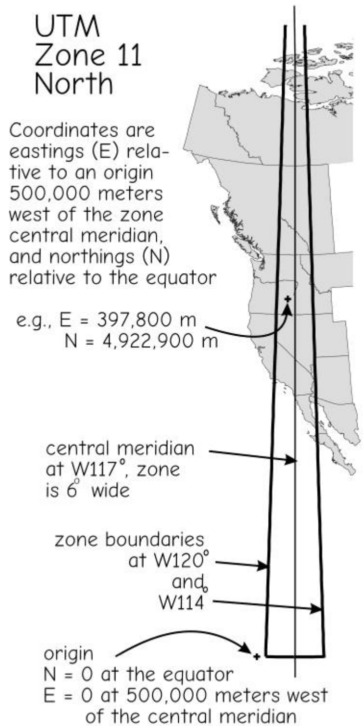
○ Universal Transverse Mercator (UTM)

- ✓ 60 zones
- ✓ Location provided in meters as **Easting** and **Northing**
- ✓ **Northing:** meters north of equator
  - In northern hemisphere, equator is assigned a value of 0 meters, add as you go up
  - In southern hemisphere, equator is assigned a value of 10,000,000 meters, subtract as you go down
- ✓ **Easting:** distance from central meridian in UTM zone
  - Central meridian is assigned a value of 500,000 meters
  - Add as you go east
  - Subtract as you go west

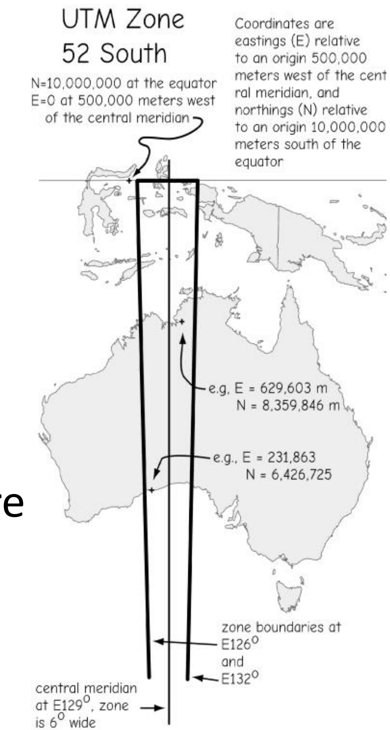
Map projection

- Universal Transverse Mercator (UTM)

Northern hemisphere

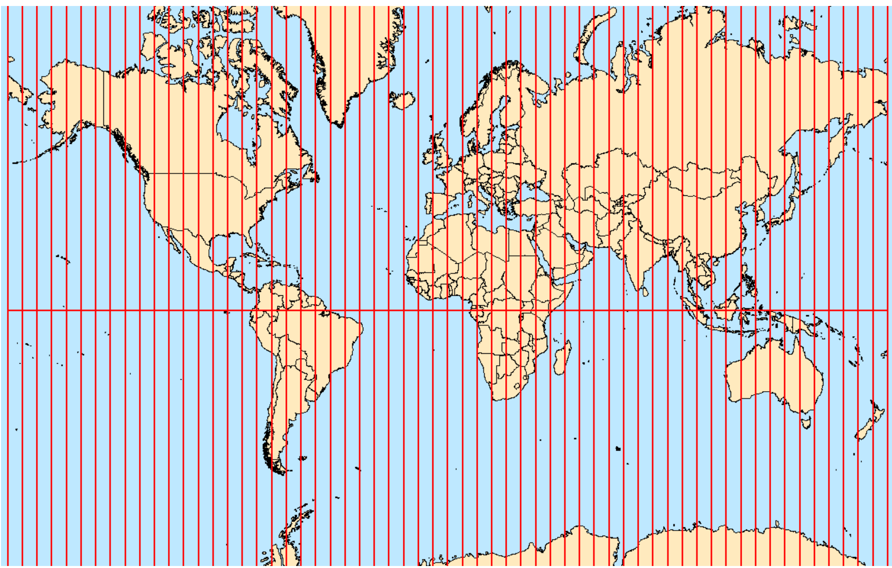


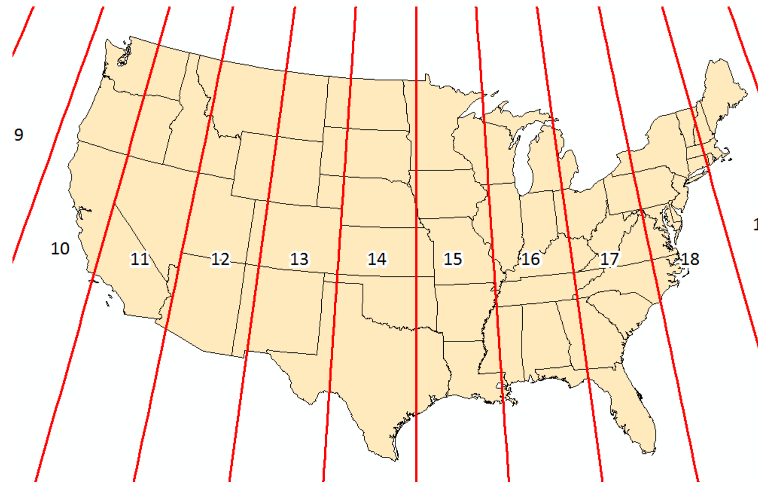
Southern hemisphere



Map projection

- Universal Transverse Mercator (UTM)



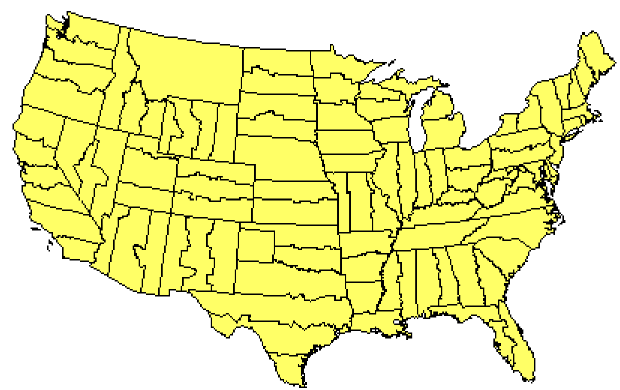


Bolstad, 2019

Map projection

○ State Plane Coordinate System

- ✓ 125 geographic zones designed for specific regions in the United States
- ✓ Zones follow county boundaries
- ✓ city- to county-scale

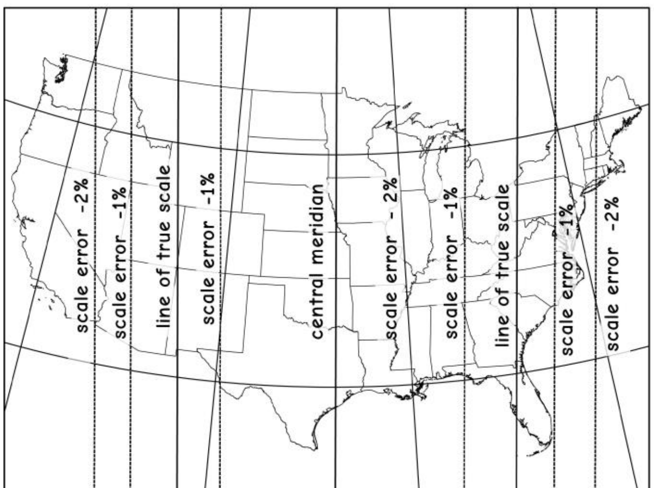


https://www.e-education.psu.edu/natureofgeoinfo/c2_p26.html

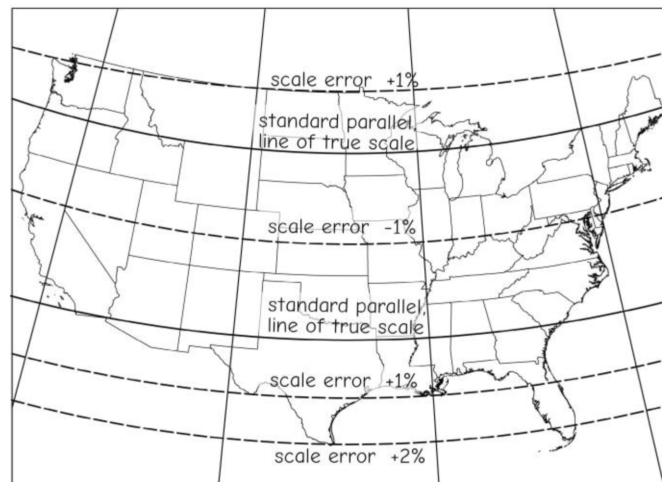
Map projection

○ State Plane Coordinate System

- ✓ Transverse Mercator projection (for states with long north-south axis)
- ✓ Lambert conformal conic projection (for states with long east-west axis)



Transverse Mercator

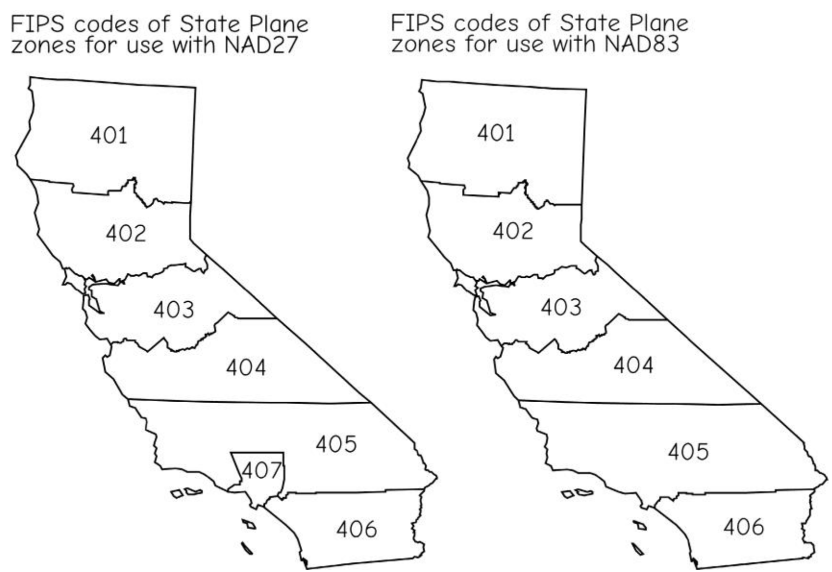


Lambert conformal conic

Bolstad, 2019

Map projection

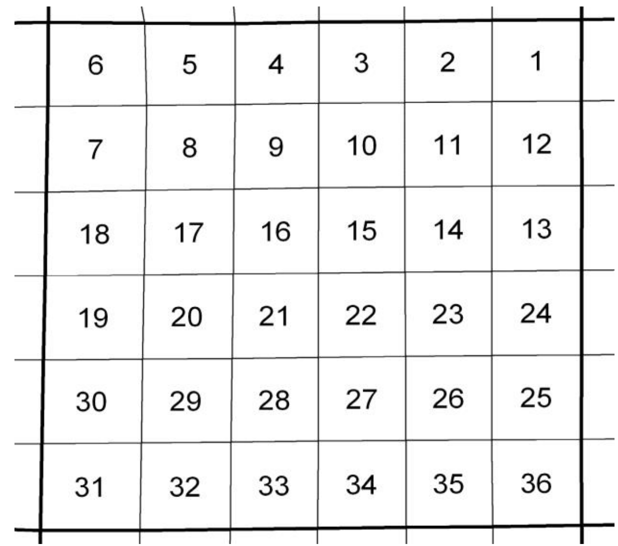
- State Plane Coordinate System
  - ✓ The zones may change!



Bolstad, 2019

Public Land Survey System

- It is not a coordinate system, but used widely for land parcels since 1800s
  - ✓ Divided lands by north-south lines, 6 miles apart
  - ✓ East-west lines were surveyed perpendicular to those north-south lines, also at 6 miles intervals
  - ✓ 30 states adopted PLSS



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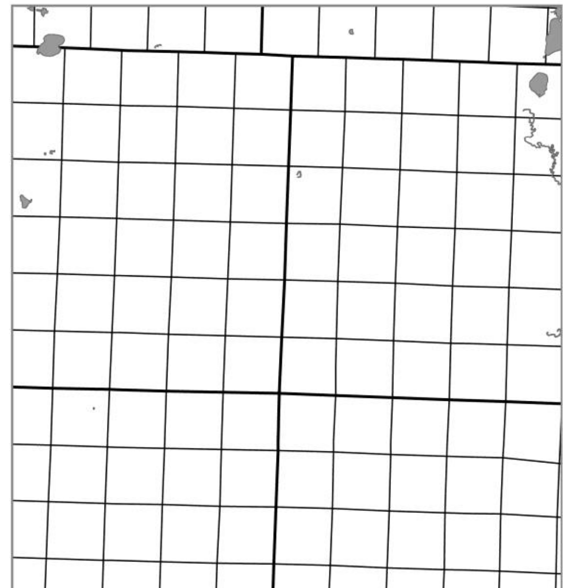
Public Land Survey System

- Place roads, powerlines, and other public services



Bolstad, 2019

Map projection

- Projection used is related to type of map and area covered by map (extent)
- Continental-scale data (North America)
  - ✓ Geographic projection (decimal degrees)
  - ✓ Lambert Conformal Conic
  - ✓ Albers Equal Area
- State-scale data (Arkansas)
  - ✓ UTM Projection, Zone 15N
- County, city-scale data
  - ✓ State Plane

Best Practice

1. Should use same projection and datum for all data
2. Use projection appropriate for scale and extent
3. Think about what map properties are maintained or distorted

Projection-on-the-Fly (ArcGIS)

- ArcGIS can apply a projection transformation on-the-fly
- So, it is not always necessary to reproject data
- Okay if you are visualizing the data
- Need to reproject if you plan to perform an analysis

Thank you!